

Axler 3.86: Matrix of Inverse Equals Inverse of Matrix

A Proof and Conceptual Commentary

Linear Algebra Done Right, 4th Edition

Context in the Text

Theorem 3.86 is the **last result stated in Chapter 3D** of *Linear Algebra Done Right*, 4th edition, appearing just before the end-of-chapter exercises. It is presented without proof, with Axler noting only that “the proof of the next result is left as an exercise.”

Chapter 3D culminates in the change-of-basis formula (3.84), which shows how the matrix of an operator transforms under a change of basis via $A = C^{-1}BC$. Theorem 3.86 is a natural coda to this: having established how matrices of operators compose and change with basis, the chapter closes by confirming that the matrix correspondence also respects inversion. It is intentionally elementary — a short exercise designed to make the reader explicitly invoke the multiplicativity machinery developed in the chapter rather than taking the result for granted.

Statement

Theorem 0.1 (3.86 — Matrix of inverse equals inverse of matrix). *Suppose $v_1, \dots, v_n$ is a basis of V and $T \in \mathcal{L}(V)$ is invertible. Then*

$$\mathcal{M}(T^{-1}) = (\mathcal{M}(T))^{-1},$$

where both matrices are with respect to the basis $v_1, \dots, v_n$.

Prerequisites

This proof relies on one named result and one direct consequence of the definition:

- **Multiplicativity of $\mathcal{M}$** (Axler 3.43): For operators $S, T \in \mathcal{L}(V)$ with respect to the same basis,

$$\mathcal{M}(ST) = \mathcal{M}(S) \mathcal{M}(T).$$

- **Matrix of the identity** (from the definition of $\mathcal{M}$): The k -th column of $\mathcal{M}(T)$ is determined by writing Tv_k as a linear combination of $v_1, \dots, v_n$. For the identity

map, $Iv_k = v_k$, so the k -th column has a 1 in position k and 0s elsewhere. Therefore $\mathcal{M}(I) = I$ (the $n \times n$ identity matrix) follows directly from the definition, without a separate theorem.

Proof

Let $B = \mathcal{M}(T)$ and $B' = \mathcal{M}(T^{-1})$, both with respect to $v_1, \dots, v_n$.

Since T is invertible, we have $TT^{-1} = I$ as operators. Applying $\mathcal{M}$ to both sides and invoking multiplicativity:

$$BB' = \mathcal{M}(T) \mathcal{M}(T^{-1}) = \mathcal{M}(TT^{-1}) = \mathcal{M}(I) = I.$$

By an identical argument using $T^{-1}T = I$:

$$B'B = \mathcal{M}(T^{-1}) \mathcal{M}(T) = \mathcal{M}(T^{-1}T) = \mathcal{M}(I) = I.$$

Since $BB' = I$ and $B'B = I$, we conclude that $B' = B^{-1}$, i.e.,

$$\mathcal{M}(T^{-1}) = (\mathcal{M}(T))^{-1}. \quad \square$$

Key Concept: Why This Exercise Exists

Key Concept. The result *feels* obvious, and it is — but only once you have identified the two bridges being crossed:

1. **Operator composition $\rightarrow$ matrix multiplication.** The map $T \mapsto \mathcal{M}(T)$ converts operator composition into matrix multiplication. This is non-trivial and was the whole motivation behind defining matrix multiplication the way it is in Chapter 3.
2. **Identity operator $\rightarrow$ identity matrix.** The map sends I (the identity *operator*) to I (the identity *matrix*). Without explicitly invoking $\mathcal{M}(I) = I$, one has not actually closed the argument.

The proof chains these two facts together: the operator equation $TT^{-1} = I$ is lifted to the matrix equation $\mathcal{M}(T) \mathcal{M}(T^{-1}) = I$ by crossing both bridges simultaneously.

Note. Algebraically, the map $T \mapsto \mathcal{M}(T)$ (with respect to a fixed basis) is a *ring isomorphism* from $\mathcal{L}(V)$ to $M_{n \times n}(F)$. Ring isomorphisms necessarily:

- send the multiplicative identity to the multiplicative identity, and
- send invertible elements to invertible elements, preserving inverses.

Theorem 3.86 is precisely the second of these properties. The “obviousness” of the result is a reflection of how well-designed the correspondence between operators and matrices is — but the formal proof still requires citing the machinery that establishes that correspondence.